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To Design ESP32-Driven System for Soil Health and Climate Adaptive Irrigation Network Using Iot

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Abstract: *This project presents an IoT-based smart irrigation system designed using ESP32 technology for efficient water management. The system monitors soil moisture, temperature, humidity, and rainfall conditions in real time. It uses ESP32, ESP32-CAM, and NodeMCU modules for data collection and control operations. Soil moisture sensors are deployed in two different farms to ensure proper irrigation. A DHT11 sensor and rain sensor help in analyzing environmental conditions accurately. The system supports both manual and automatic modes for flexible operation. A Telegram bot is developed as a user interface for remote monitoring and control. Users can operate pumps, view system status, and capture field images. The ESP32-CAM provides real-time visual monitoring of the field. The system helps in reducing water wastage and improving irrigation efficiency. Overall, it offers a cost-effective and reliable solution for smart agriculture.*

Key Word: *IoT Sensors,Smart Irrigation, ESP32, NodeMCU, Soil Moisture Sensor, DHT11, Rain Sensor, Telegram Bot, Automation, Prcision Agriculture.*

I. INTRODUCTION

Agriculture is a vital sector that supports food production and economic stability across the world. However, traditional irrigation practices rely heavily on manual effort and fixed schedules, which often lead to inefficient use of water resources. Farmers frequently face challenges in monitoring soil conditions and environmental factors such as temperature, humidity, and rainfall. This lack of real-time information can result in overwatering or insufficient irrigation, affecting crop health and yield. With increasing water scarcity and unpredictable climate conditions, there is a growing need for smarter and more efficient irrigation methods. The integration of modern technologies in agriculture can help address these issues effectively. Internet of Things (IoT) has emerged as a powerful solution for automating and optimizing agricultural processes. It allows devices and sensors to communicate and share data over the internet. This enables continuous monitoring and control of field conditions. As a result, farmers can make better decisions and improve productivity. Therefore, adopting IoT-based systems can significantly enhance the efficiency and sustainability of farming practices.

In this project, an IoT-based smart irrigation system is developed using ESP32, ESP32-CAM, and NodeMCU modules. The system is designed to monitor soil moisture, temperature, humidity, and rainfall conditions in real time using various sensors. The collected data is processed to determine the need for irrigation, ensuring that water is supplied only when required. The system supports both manual and automatic modes, providing flexibility to the user. A Telegram bot is used as an interface for remote monitoring and control, allowing users to operate pumps and check system status from anywhere. Additionally, the ESP32-CAM module enables real-time visual monitoring of the field. This helps users observe crop conditions without being physically present. The system also provides updates such as soil moisture levels, weather conditions, and motor status. By reducing water wastage and improving irrigation accuracy, the system contributes to better crop management. It is designed to be cost-effective and easy to implement for farmers. Overall, the project demonstrates a practical approach to smart and sustainable agriculture using IoT technology.

II.PROBLEM STATEMENT

To Design ESP32-Driven System For Soil Health And Climate Adaptive Irrigation Network Using Iot

III.OBJECTIVES

- To design an IoT-based smart irrigation system that monitors soil moisture, temperature, and humidity in real-time.
- To automatically control water supply using a 5V water pump and solenoid valve based on sensor readings.
- To enable remote monitoring and manual control of irrigation through the Blynk mobile application.

- To optimize water usage and prevent over-irrigation, promoting sustainable agriculture practices.
- To provide a cost-effective and scalable solution for efficient soil and crop management in modern farming.

IV.LITERATURE REVIEW

A. Sharma et al. (2025) in YL-83 Rain Sensor for Smart Irrigation Systems. The YL-83 Rain Sensor, produced by Keyes Technology, is commonly used in automated irrigation and weather monitoring systems. It operates using a conductive plate that detects the presence of water droplets. Research shows that it provides quick response for rainfall detection. This makes it useful for automatic irrigation cut-off systems. However, it cannot measure rainfall intensity quantitatively. The sensor is low-cost and easy to integrate with microcontrollers. It consumes less power, making it ideal for IoT applications. Its simple working mechanism ensures reliability in basic rain detection. Overall, it is suitable for real-time rainfall detection in smart farming systems.

ZvinodasheFeresu et al. (2022) in DHT11 Temperature and Humidity Sensor for IoT Monitoring Systems. The DHT11 Temperature and Humidity Sensor, manufactured by Aosong Electronics, is widely used for environmental monitoring in IoT systems. It uses a capacitive humidity sensor and a thermistor for temperature measurement. Studies show that it provides acceptable accuracy for small-scale applications. The sensor gives digital output, making it easy to interface with microcontrollers. It has low power consumption and simple design. However, its response time is slower compared to advanced sensors. It is suitable for basic monitoring rather than high-precision applications. Its low cost makes it ideal for student projects. Overall, it is reliable for basic temperature and humidity measurement.

L. Yu et al. (2021) in Robodo SEN6 Soil Moisture Sensor for Smart Irrigation Systems. The Robodo SEN6 Soil Moisture Sensor, developed by Robodo Technologies, is widely used in IoT-based agricultural systems. It works on the principle of electrical resistance variation between probes to determine soil moisture levels. Studies indicate that resistive sensors like SEN6 are cost-effective and suitable for small-scale applications. The sensor provides real-time data that helps in efficient irrigation management. However, its readings can be affected by soil salinity and long-term corrosion. Despite these limitations, it is easy to interface with microcontrollers such as ESP32. The sensor requires minimal setup and calibration. Its affordability makes it popular among student and prototype projects. Overall, it is suitable for basic soil moisture monitoring systems.

Espressif Systems (2021) in ESP32-CAM Module for IoT-Based Monitoring Systems. The ESP32-CAM module, developed by Espressif Systems, is a compact camera module used in IoT-based monitoring systems. It combines a microcontroller with a camera for image capture and transmission. Research shows that it is effective for real-time surveillance and remote monitoring. It supports WiFi communication, allowing images to be sent over the internet. The module is low-cost and requires minimal external components. It is suitable for applications like smart agriculture monitoring. However, it requires proper power supply for stable operation. Its small size makes it easy to install in different environments. Overall, it is useful for visual monitoring in IoT systems.

Espressif Systems (2020) in ESP32 Microcontroller for IoT-Based Systems. The ESP32 microcontroller, developed by Espressif Systems, is widely used in IoT-based systems due to its built-in WiFi and Bluetooth capabilities. It provides high processing power and supports multiple sensors and communication protocols. Research shows that ESP32 is efficient for real-time data processing and remote monitoring. It can handle multiple tasks simultaneously, making it suitable for complex IoT systems. The controller is compatible with Arduino IDE, making development easier. It also supports low power modes for energy efficiency. Its cost-effectiveness makes it popular among developers. ESP32 plays a key role in controlling sensors and communication. Overall, it is a powerful and flexible IoT controller.

Espressif Systems (2019) in NodeMCU (ESP8266) for IoT-Based Automation Systems. The NodeMCU, based on the ESP8266 module, is a popular microcontroller used in IoT applications. It provides built-in WiFi connectivity for wireless communication. Studies show that NodeMCU is efficient for remote control and automation systems. It can easily interface with relay modules to control electrical devices. The board is programmable using Arduino IDE, making it user-friendly. It is cost-effective and widely available. Its compact design allows easy integration into projects. However, it has limited processing power compared to ESP32. Overall, it is suitable for controlling devices like motors in smart irrigation systems.

V.SOFTWARE/HARDWARE REQUIREMENTS

Software Requirements

The software components required for the development and operation of the proposed system are as follows:

- Arduino IDE

Used for writing, compiling, and uploading code to ESP32, ESP32-CAM, and NodeMCU modules.

- Telegram Application

provides a user interface for remote monitoring and control of the irrigation system through a bot.

- Telegram Bot API

Enables communication between the user and the system for sending commands and receiving updates.

- Embedded C/C++ Programming

Used for developing the control logic and sensor data processing in microcontrollers.

- WiFi Connectivity Support

required for enabling communication between system components and remote access via the internet.

Hardware Requirements

The hardware components used in the proposed system are listed below:

- ESP32 Microcontroller

Acts as the main processing unit for collecting sensor data and controlling system operations.

- ESP32-CAM Module

Used for capturing real-time images of the agricultural field for visual monitoring.

- NodeMCU (ESP8266) Modules (2 Units)

Used for controlling water pumps through relay modules for two separate farms.

- Soil Moisture Sensors (SEN6) (2 Units)

Measure soil moisture levels to determine irrigation requirements.

- DHT11 Sensor

Measures temperature and humidity of the environment.

- YL-83 Rain Sensor

Detects rainfall conditions to prevent unnecessary irrigation.

- Relay Modules (2-Channel)

Used to control ON/OFF operation of water pumps.

- Water Pumps (12V)

Used for supplying water to the fields based on system control.

- Power Supply (5V Adapter and 12V Source)

Provides required power to microcontrollers and pumps.

- Level Shifter Module

Used for voltage level conversion between different components.

VI.SYSTEM ARCHITECTURE

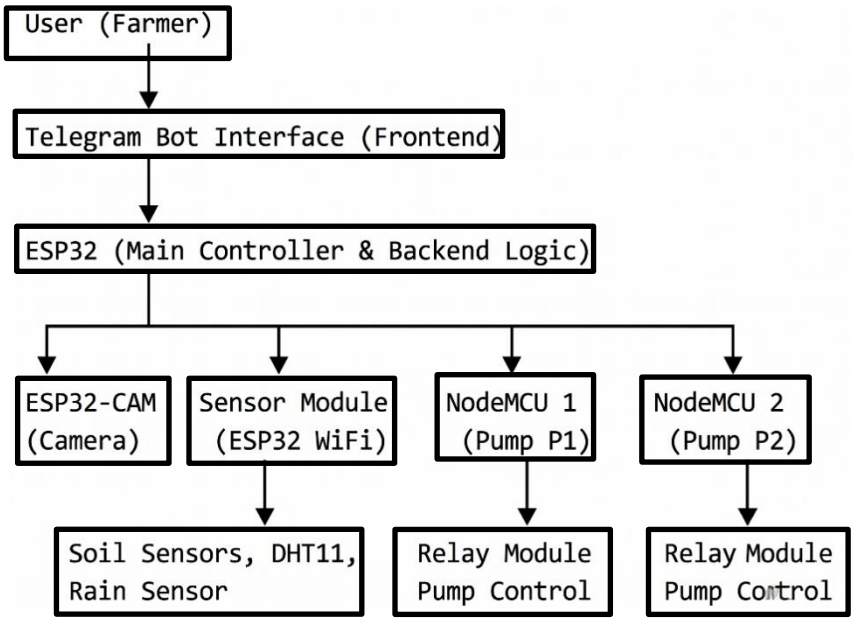


Fig1: System Architecture

- **Level 0 – User Level**

The farmer interacts with the system using the Telegram bot.

- **Level 1 – Interface Level**

The Telegram Bot Interface acts as the frontend to receive commands and display system responses.

- **Level 2 –**Control Level (ESP32)

The ESP32 processes user commands and acts as the main controller of the system.

- **Level 3 –**Module Level

The ESP32 distributes tasks to ESP32-CAM, Sensor Module, NodeMCU 1, and NodeMCU 2.

- **Level 4 –** Hardware Level

Sensors collect data and relay modules control the pumps.

The system is structured in five levels, starting from the user interface and ending at the hardware layer. The farmer interacts with the system through the Telegram bot, which forwards commands to the ESP32 controller. The ESP32 processes the data and distributes tasks to different modules such as sensors, camera, and pump controllers. Finally, sensors provide real-time data and relay modules execute irrigation control, completing the system workflow.

VII.DATA FLOW

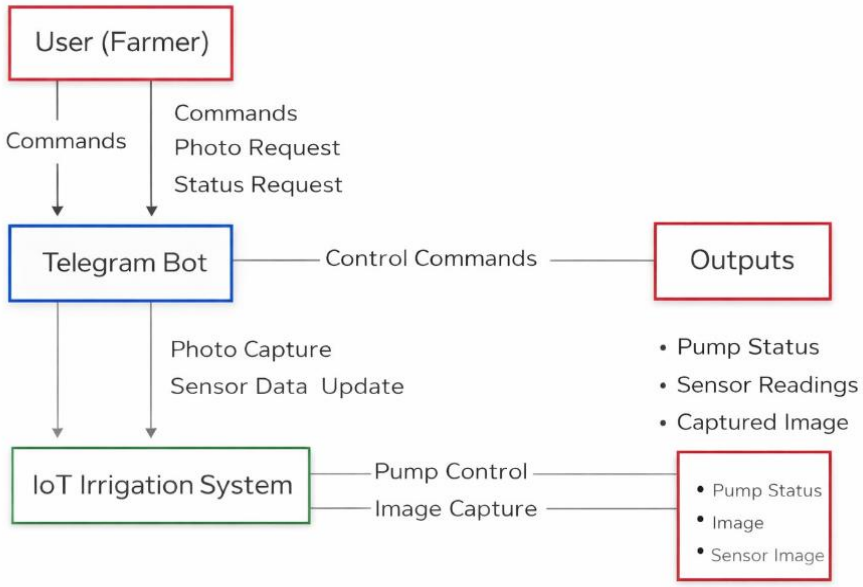


Fig 2: Data Flow

- **User (Farmer)**

The farmer sends commands such as pump control, photo request, and status request through the Telegram bot.

- **Telegram Bot**

The bot receives user inputs, processes them, and forwards the commands to the IoT irrigation system.

- **IoT Irrigation System**

The system processes commands, collects sensor data, and performs actions like pump control and image capture.

- **Sensor Data Flow**

Sensor data such as soil moisture, temperature, humidity, and rainfall is continuously collected and sent to the system.

- **Control Command Flow**

Commands from the user are converted into control signals for pump operation and system actions.

- **Output Flow**

The system sends results including pump status, sensor readings, and captured images back to the Telegram bot.

- **Feedback to User**

The Telegram bot delivers processed outputs to the farmer for real-time monitoring and decision-making.

The data flow diagram represents the communication between the farmer, Telegram bot, and IoT irrigation system. The farmer sends commands such as pump control, photo request, and status request through the Telegram bot. The system processes these commands, collects real-time sensor data, and performs actions like pump control and image capture. The outputs, including pump status, sensor readings, and captured images, are sent back to the user for monitoring and control.

VIII.UML DIAGRAM

The UML diagram represents the overall structure and workflow of the IoT-based smart irrigation system. It includes use case, class, sequence, and activity diagrams to explain different functionalities of the system. The use case diagram shows interactions between the farmer and the system through the Telegram bot. The class diagram represents relationships between modules such as ESP32, NodeMCU, sensors, and ESP32-CAM. The sequence and activity diagrams illustrate the flow of commands, data processing, pump control, and communication between components for efficient irrigation management.

- Works best in areas with stable internet and WiFi connectivity for real-time monitoring and control.
- Performs efficiently in small and medium-scale farms requiring automated irrigation management.

Types of UML Diagrams

1. **Use Case Diagram**

- The use case diagram represents interactions between the user and the IoT irrigation system functionalities.
- It shows operations such as pump control, sensor monitoring, status requests, and image capture.

2. **Clase Diagram**

- The class diagram shows the structure and relationships between system components and modules.
- It represents interactions between sensors, controllers, and irrigation control units.

3. Sequence Diagram

- The sequence diagram shows the flow of communication between the user, Telegram bot, ESP32, and hardware modules.
- It represents the step-by-step execution of commands, sensor data processing, and pump control operations.

4. Activity Diagram

- The activity diagram represents the workflow of the smart irrigation system from data collection to pump operation.
- It shows the sequence of activities involved in monitoring, decision-making, and irrigation control.

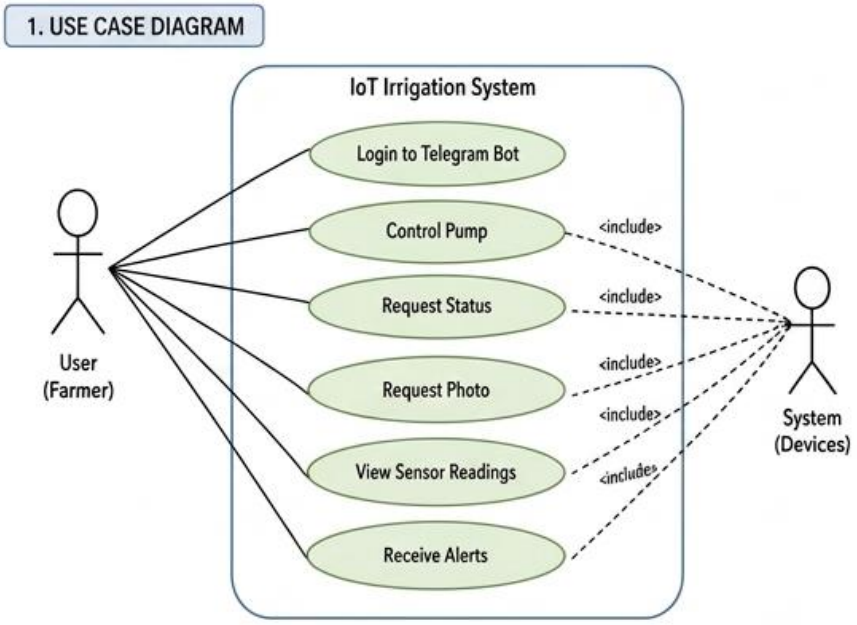


Fig 3: Use Case Diagram

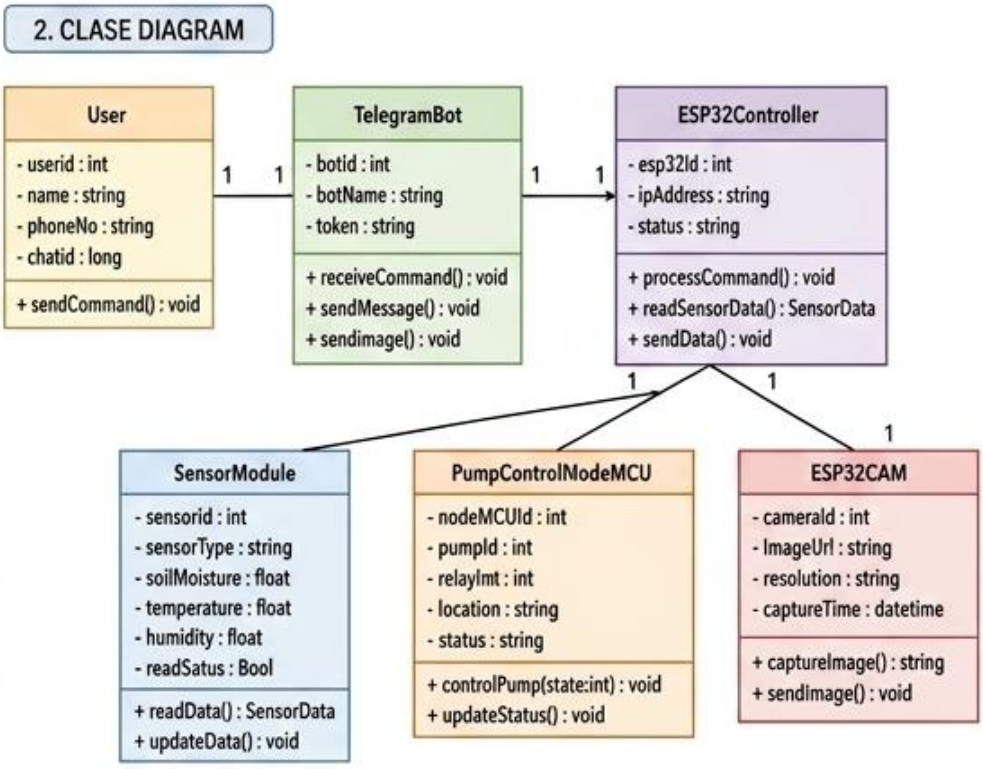


Fig 4 :Clase Diagram

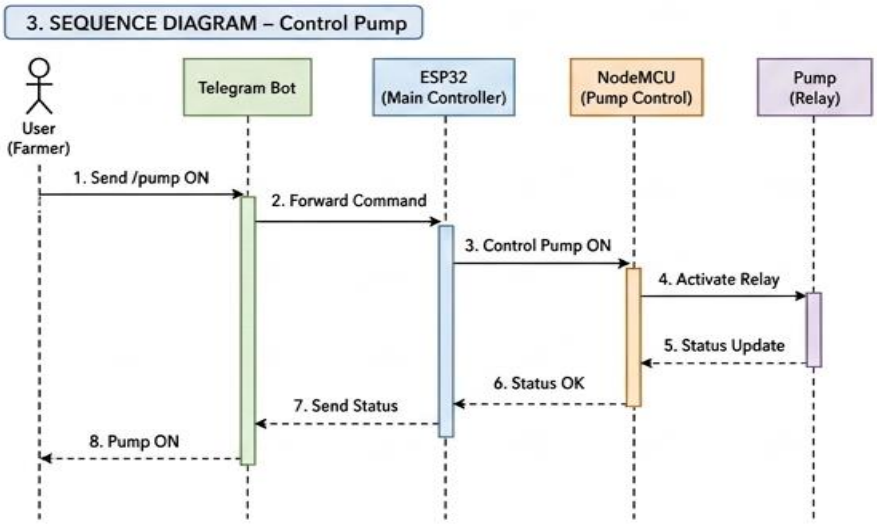


Fig 5: Sequence Diagram

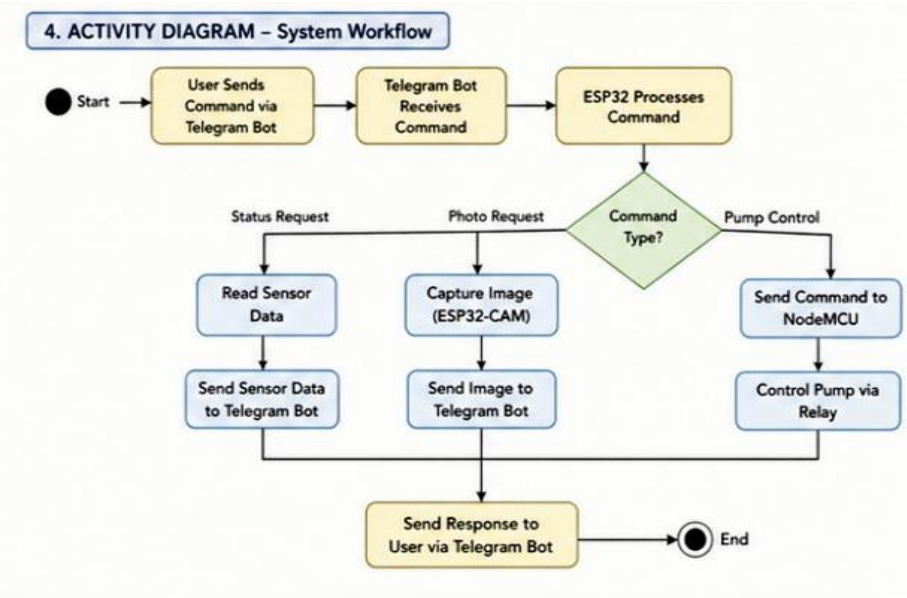


Fig 6: Activity Diagram

IX.ENTITY RELATIONSHIP DIAGRAM

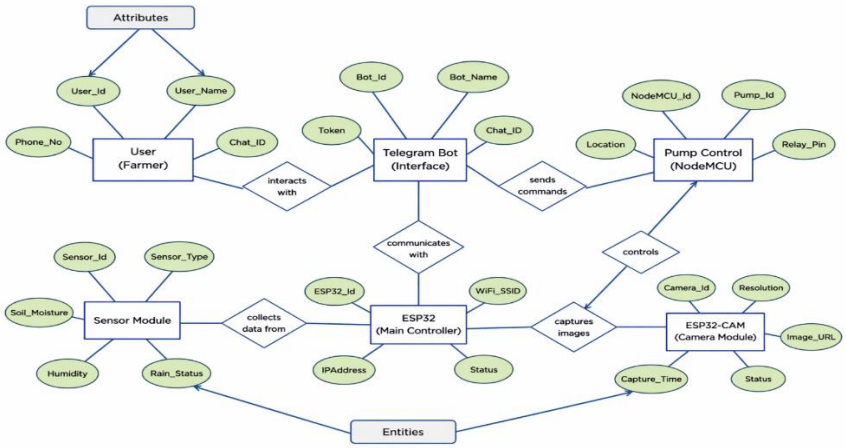


Fig 7: ER Diagram

- **User (Farmer)**

The user interacts with the system through the Telegram bot to control and monitor irrigation.

- **Telegram Bot**

Acts as an interface that receives user commands and sends system responses.

- **ESP32 (Main Controller)**

Processes sensor data and manages overall system operations and decision-making.

- **Sensor Module**

Collects real-time environmental data such as soil moisture, temperature, humidity, and rainfall.

- **NodeMCU (Motor Control)**

Controls water pumps using relay modules based on commands from ESP32.

- **ESP32-CAM**

Captures real-time images of the field and sends them to the user.

- **Outputs**

Displays system results including pump status, sensor readings, and captured images.

X.RESULT

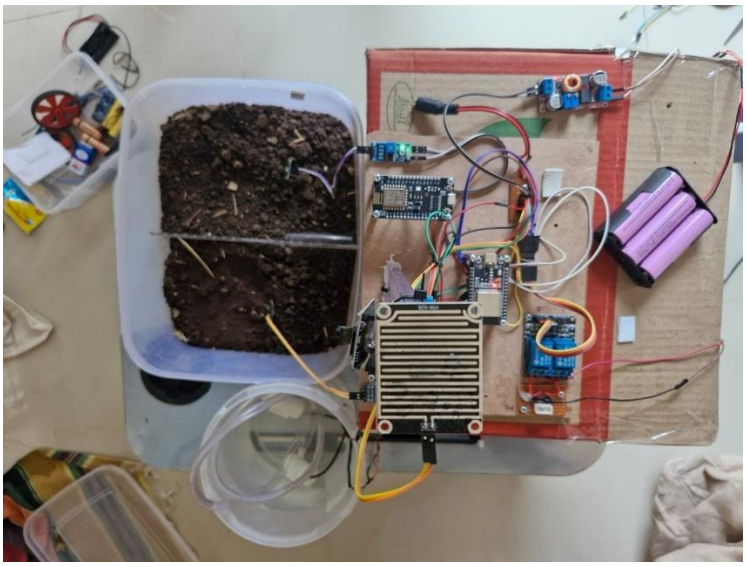


Fig 8: Project Deployment Setup

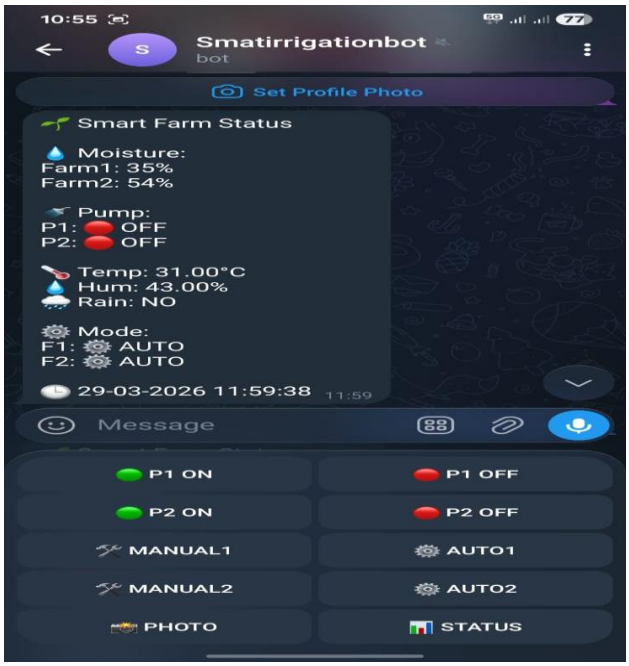


Fig 9: Telegram Operating Interface

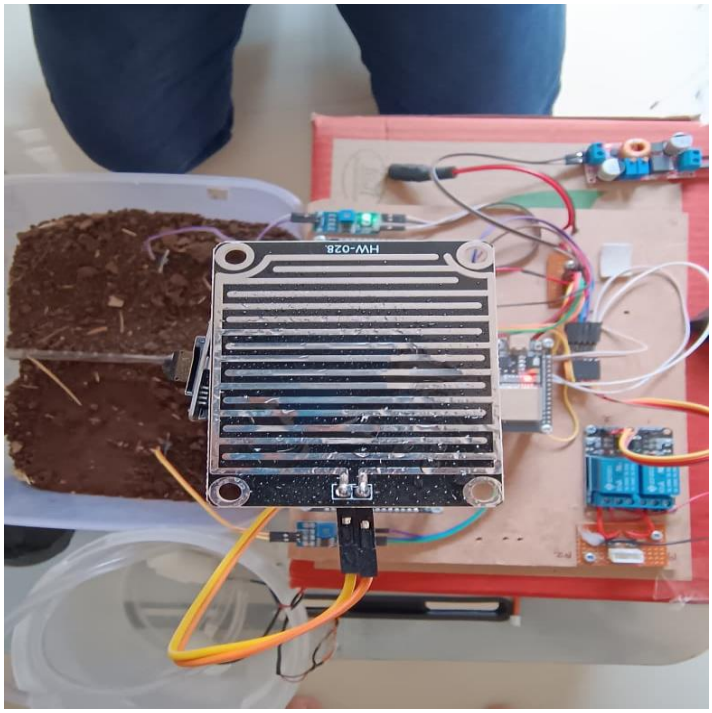


Fig 10: Rain Sensor

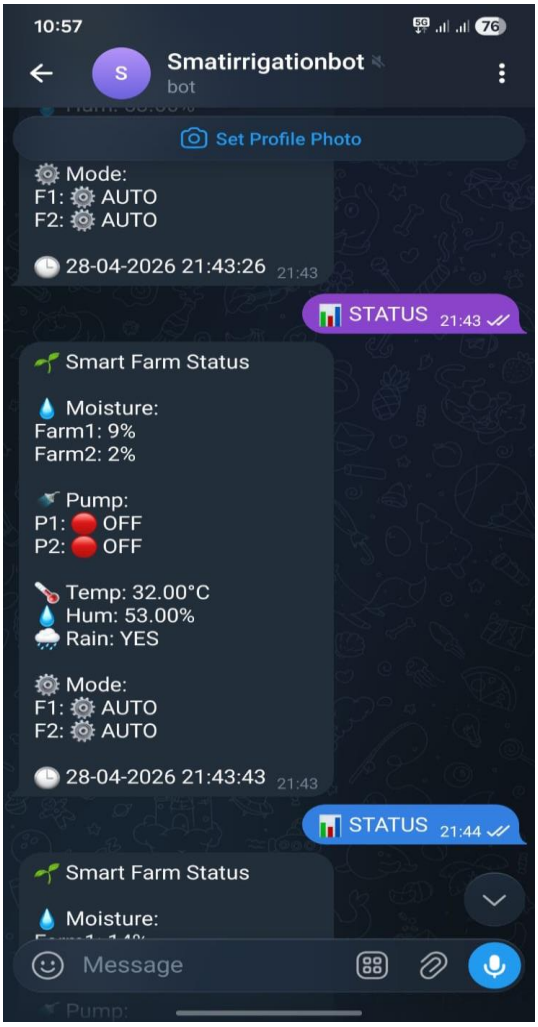


Fig 11: Rain Status

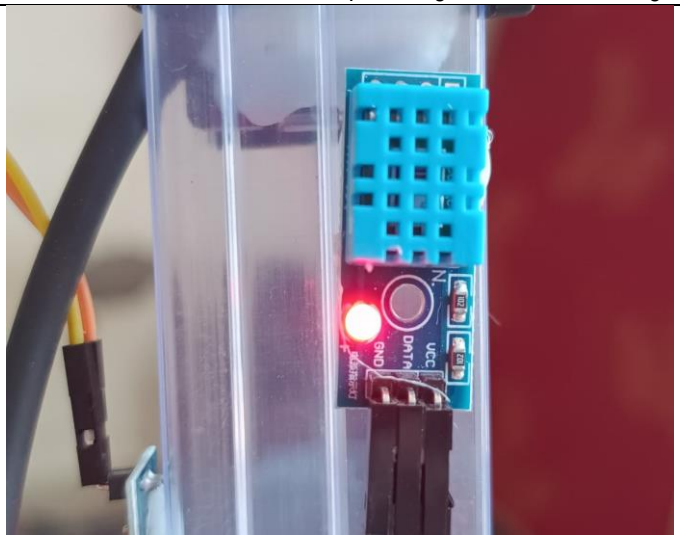


Fig 12 : Temperature And Humidity Sensor

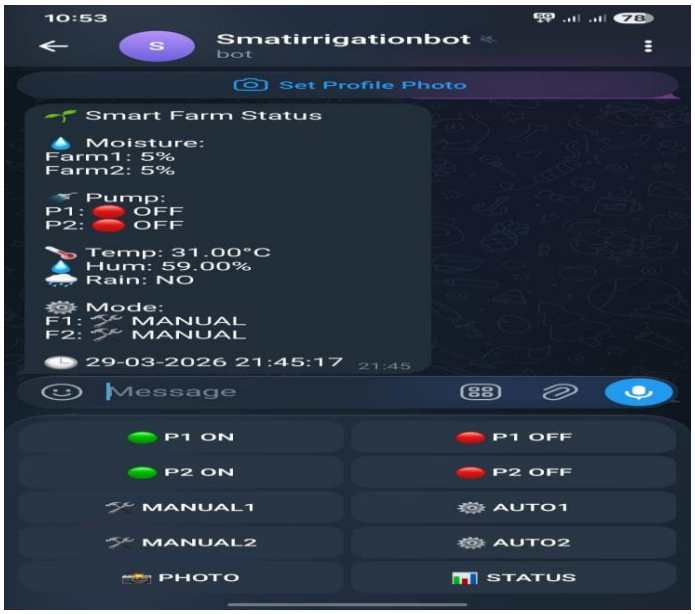


Fig 13: Temperature And Humidity Status

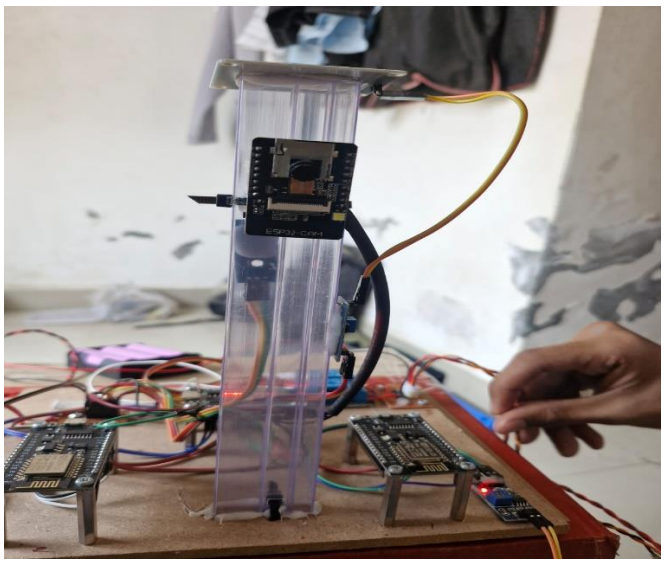


Fig 14: Esp-32 Cam Sensor

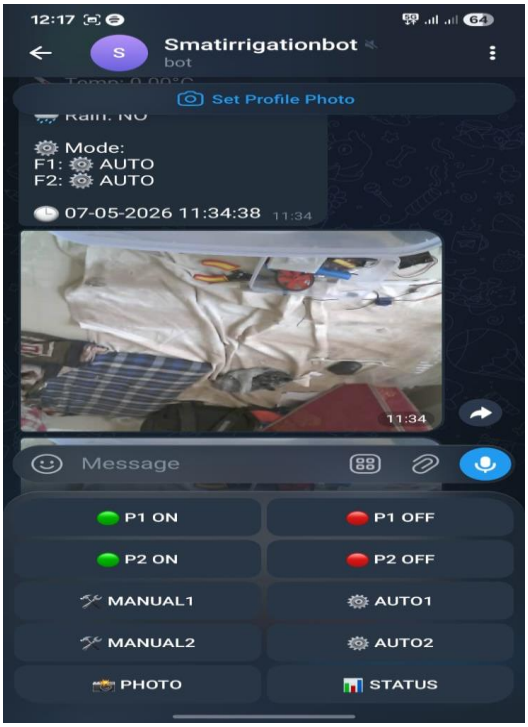


Fig 15: Camera Status

XI.ADVANTAGES

- **Efficient Water Management**: The system supplies water only when required, reducing wastage.
- **Automation of Irrigation**: Irrigation is controlled automatically based on real-time sensor data.
- **Remote Monitoring and Control**: Users can monitor and operate the system from anywhere using the Telegram bot.
- **Improved Crop Productivity**: Proper irrigation helps in maintaining optimal conditions for better crop growth.
- **Reduced Human Effort**: The system minimizes manual work involved in irrigation management.
- **Cost-Effective Solution**: Uses affordable components, making it suitable for small and medium-scale farmers.

XII.DISADVANTAGES

- **Dependence on Internet Connectivity**: The system requires a stable internet connection for proper operation.
- **Initial Setup Cost**: Initial installation of hardware components may be costly for some farmers.
- **Technical Complexity**: Users may need basic technical knowledge to install and maintain the system.
- **Power Supply Requirement**: Continuous power is required for sensors, controllers, and communication.
- **Sensor Accuracy Limitations**: Inaccurate sensor readings can affect system performance and decisions.
- **Maintenance Requirement**: Regular maintenance is needed to ensure proper functioning of hardware components.

XIII.APPLICATION

- **Smart Agriculture**: Used for efficient irrigation management in modern farming.
- **Greenhouse Monitoring**: Helps in controlling environmental conditions inside greenhouses.
- **Garden Irrigation Systems**: Automates watering for home gardens and landscaping.
- **Precision Farming:** Supports accurate irrigation based on real-time data.
- **Water Resource Management:** Helps in conserving water by optimizing usage.

XIV.LIMITATION

- **Dependence on Internet Connectivity:** The system requires a stable internet connection; without it, remote control and monitoring will not work properly.
- **Sensor Accuracy Issues:** Low-cost sensors may give inaccurate readings due to environmental conditions, affecting system decisions.
- **Power Dependency:** Continuous power supply is required; any power failure stops system operation.
- **Basic Automation Logic:** The system uses simple threshold-based control and lacks advanced intelligent decision-making.
- **Maintenance Requirement:** Hardware components like sensors and pumps may require regular maintenance over time.

XV.CONCLUSION

This project demonstrates an IoT-based smart irrigation system for efficient water management using ESP32, ESP32-CAM, and NodeMCU. It collects real-time data from soil moisture, temperature, humidity, and rain sensors to support accurate irrigation decisions. The system operates in both manual and automatic modes, allowing flexible control through a Telegram bot interface. It also provides visual monitoring using ESP32-CAM for better field observation. The solution reduces water wastage, saves time, and improves crop productivity. Overall, it offers a cost-effective, reliable, and scalable approach for modern smart agriculture.

XVI.FUTURE SCOPE

The proposed system can be further enhanced by integrating a dedicated mobile application to provide a better user interface and improved control. Sensor data can be stored on cloud platforms for long-term monitoring and advanced data analysis. Artificial Intelligence techniques can be applied to predict irrigation requirements based on historical data and environmental conditions. The system can also be powered using solar energy to reduce dependency on external power sources. It can be expanded to support multiple farms and larger agricultural areas for wider implementation. Additional sensors such as pH, nutrient, and water level sensors can be integrated to provide more detailed soil and crop information. Voice control features can be added to simplify system operation. Real-time weather forecast data can be used to further optimize irrigation decisions. A web-based monitoring dashboard can be developed for detailed visualization and control. Moreover, advanced security features such as authentication and encryption can be implemented to ensure safe and reliable system operation.

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